

Your thesis title

Your name

DEPARTMENT OF PHYSICS AND ASTRONOMY, BATES COLLEGE, LEWISTON, ME 04240

Your thesis title

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Presented to the Department of Physics and Astronomy

Bates College

in partial fulfillment of the requirements for the

Degree of Bachelor of Arts

by

Your name

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Acknowledgments

Mom, Dad, your roommate, maybe even your thesis advisor – give credit where credit’s due! For my part, I’d like to thank the Math department for making such a nice L^AT_EX template that we could use as a base for our Physics template.

Introduction

You can, if you wish, write an introduction.

CHAPTER 1

Theory Notes 1

1. HITRAN linefunction model

2. Writing equations in L^AT_EX

It is beautifully easy to write equations, whether they occur inline with the text $\Psi(x, t) = Ae^{i(kx - \omega t)}$ or as a separate line

$$(1.1) \quad \Psi(x, t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-iE_n t/\hbar}.$$

To continue a paragraph after an equation, make sure that there's no blank line between the equation and your next line of text. Equations that are in a separate line will be numbered, and it's easy to refer to Eq. 1.2 by labelling your equations

$$(1.2) \quad i\hbar \frac{\partial}{\partial t} \Psi(x, t) = \hat{H} \Psi(x, t).$$

Are you getting tired of writing `begin{eqnarray}` and `end{eqnarray}` yet?¹ You can simplify your life by defining abbreviated symbols for commands:

$$(1.3) \quad \langle x | \Psi(t) \rangle = \Psi(x, t).$$

For simplicity, I'd suggest putting all of your new commands in a separate include file.

When writing a derivation with multiple steps, it is often useful to use an equation array

$$\begin{aligned} \int_0^{2\pi} \sin^2 x dx &= \int_0^{2\pi} \frac{1}{2} (1 - \cos 2x) dx \\ &= \frac{1}{2} \left(x - \frac{\sin 2x}{2} \right) \Big|_0^{2\pi} \\ &= \pi. \end{aligned}$$

Note that you can suppress equation numbering, which is often desirable in derivations!

The syntax for equation arrays is similar to the syntax for arrays within equations. For example,

$$(1.4) \quad \hat{H} = \begin{pmatrix} \Delta & \Omega & 0 \\ \Omega^* & 0 & \kappa \\ 0 & \kappa^* & -\Delta \end{pmatrix}$$

There are many more complicated things you can do with equations, but this should be enough to get started!

¹Notice how I made a `}` by putting a backslash in front of the symbol. The curly brace has special meaning in L^AT_EX, so if you want to make one, you have to let the program know that you want the symbol.

3. Tables, lists, etc... etc...

If you want to make a table, you'll need to specify the justification for each of the columns; T_EX will automatically calculate the column widths for you.

Name	Age	Town	Time
Joe Schmoe	23	Rumford	1:04
Mary Q. Contrary	44	Yarmouth	1:12
Baxter Boulevard	32	Portland	1:42
Lass Plaice	60	Lewiston	1:57

TABLE 1. Not too many people showed up for the first, and probably last, annual Androscoggin swimming race.

You might also, for example, want to

- (1) make lists
- (2) number your lists
 - use bullet points
 - find better items to put in your bullet points

4. Figures

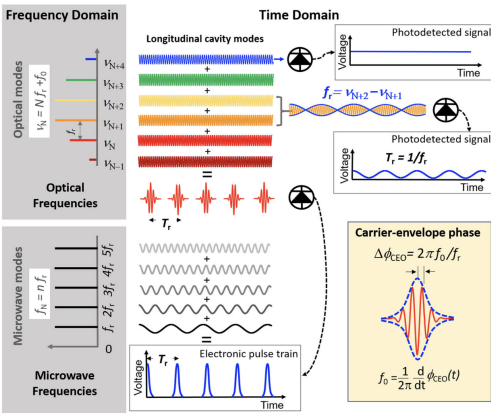


FIGURE 1.1. This is a very bad figure.

5. References

test citation [1]

CHAPTER 2

Some More Things you can do in $\text{\LaTeX}$

WOOOOO

Appendix A

Long derivations, marginally relevant details, and the like can be relegated to your appendix, which unlike the mammalian counterpart will never require removal.

Bibliography

- [1] D.E. Chang et al. “Quantum optics with surface plasmons”. In: *Physical Review Letters* 97 (2006), p. 053002.